

A sign error at critical structural damping: counterexample and corrected Mittag–Leffler kernels

Full correction to the $\mu = 2$ formulas in arXiv:2212.10463v2

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Result. In the critical case $\mu = 2$, two kernels used in Theorems 6.3 and 6.4 of Faustino–Marques [1] have the wrong sign. For every $1/2 < \alpha < 1$, their stated hyperbolic solution with nonzero initial velocity fails the prescribed initial condition. The error comes from an incorrect recurrence for the Prabhakar function. We prove the correct recurrence, give a direct counterexample, and replace both affected kernels. The initial-displacement kernel is already correct.

1 The printed identity

The three-parameter Mittag–Leffler (Prabhakar) function is

$$E_{\alpha,\beta}^{\gamma}(w) = \sum_{k=0}^{\infty} \frac{(\gamma)_k}{k!} \frac{w^k}{\Gamma(\alpha k + \beta)}.$$

The source’s Lemma 4.3(iii) prints

$$\alpha E_{\alpha,\beta}^2(w) = E_{\alpha,\beta-1}(w) - (1 + \alpha - \beta)E_{\alpha,\beta}(w). \quad (1)$$

This identity is then used in the repeated-root formulas on pages 23, 25, and 27.

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Lemma 4.3 (cf. [24], p. 99). *Let $z \in \mathbb{C}$ be given.*

(i) *If $\alpha, \beta, \gamma \in \mathbb{C}$ are such that $\Re(\alpha) > 0$, $\Re(\beta) > 0$, $\Re(\beta - \alpha) > 0$, then*

$$z E_{\alpha,\beta}^{\gamma}(z) = E_{\alpha,\beta-\alpha}^{\gamma}(z) - E_{\alpha,\beta-\alpha}^{\gamma-1}(z).$$

(ii) *If $\alpha, \beta, \gamma \in \mathbb{C}$ are such that $\Re(\alpha) > 0$, $\Re(\beta) > 0$, $\alpha - \beta \notin \mathbb{N}_0$, then*

$$z E_{\alpha,\beta}(z) = E_{\alpha,\beta-\alpha}(z) - \frac{1}{\Gamma(\beta - \alpha)}.$$

(iii) *If $\alpha, \beta \in \mathbb{C}$ are such that $\Re(\alpha) > 0$, $\Re(\beta) > 1$, then*

$$\alpha E_{\alpha,\beta}^2(z) = E_{\alpha,\beta-1}(z) - (1 + \alpha - \beta)E_{\alpha,\beta}(z).$$

4.2. Fourier analysis, function spaces and associated operators. Let us now denote by $\mathcal{S}(\mathbb{R}^n)$ the Schwartz space over $\mathbb{R}^n$ and by $\mathcal{S}'(\mathbb{R}^n)$ the *space of tempered*

Figure 1: Lemma 4.3(iii) in arXiv:2212.10463v2, page 9. The displayed minus sign must be a plus sign.

There is already a zero-order check: at $w = 0$, the left side of (1) is $\alpha/\Gamma(\beta)$, whereas the printed right side is $(2\beta - \alpha - 2)/\Gamma(\beta)$. These agree only in the exceptional case $\beta = \alpha + 1$.

2 Correct recurrence

Theorem 1 (Prabhakar reduction). *For $\alpha > 0$ and $\beta \in \mathbb{R}$,*

$$\boxed{\alpha E_{\alpha,\beta}^2(w) = E_{\alpha,\beta-1}(w) + (1 + \alpha - \beta)E_{\alpha,\beta}(w).} \quad (2)$$

Consequently, the difference “printed minus correct” is

$$\frac{2(\beta - \alpha - 1)}{\alpha} E_{\alpha,\beta}(w) \quad (3)$$

after division by α .

Proof. Since $(2)_k/k! = k + 1$,

$$E_{\alpha,\beta}^2(w) = \sum_{k \geq 0} (k + 1) \frac{w^k}{\Gamma(\alpha k + \beta)}.$$

On the other hand,

$$E_{\alpha,\beta-1}(w) = \sum_{k \geq 0} (\alpha k + \beta - 1) \frac{w^k}{\Gamma(\alpha k + \beta)}.$$

Adding $(1 + \alpha - \beta)E_{\alpha,\beta}(w)$ makes the coefficient $\alpha(k + 1)$ term by term, proving (2). Subtracting the two alternatives gives (3). $\square$

The same formula also follows by taking the divided-difference limit

$$\lim_{\lambda_+, \lambda_- \rightarrow 1} \frac{\lambda_+ E_{\alpha,\beta}(-\lambda_+ z) - \lambda_- E_{\alpha,\beta}(-\lambda_- z)}{\lambda_+ - \lambda_-}.$$

Thus the apparent pole at the coalescing characteristic roots is removable, and its value has the plus sign in (2).

3 Correct critical kernels

Write $a = |\xi|^\sigma$ and $z = at^\alpha$. At $\mu = 2$ the transformed denominator is $(s^\alpha + a)^2$. The source correctly obtains the Prabhakar forms before replacing them by two-parameter functions:

$$\mathcal{L}^{-1} \left[\frac{s^{2\alpha-\beta}}{(s^\alpha + a)^2} \right] = t^{\beta-1} E_{\alpha,\beta}^2(-z), \quad (4)$$

$$\mathcal{L}^{-1} \left[\frac{s^{2\alpha-2}}{(s^\alpha + a)^2} \right] = t E_{\alpha,2}^2(-z). \quad (5)$$

Indeed the general identity is

$$\mathcal{L}\{t^{\beta-1} E_{\alpha,\beta}^2(-at^\alpha)\}(s) = \frac{s^{2\alpha-\beta}}{(s^\alpha + a)^2}.$$

Proposition 1 (Replacement formulas). *In Notations 6.1–6.2 and equations (6.10), (6.18) of [1], replace the two critical multipliers by*

$$m_{\text{crit}}(t, \xi) = \frac{t^{\beta-1}}{\alpha} \left(E_{\alpha,\beta-1}(-z) + (1 + \alpha - \beta) E_{\alpha,\beta}(-z) \right), \quad (6)$$

$$j_{\text{crit}}(t, \xi) = \frac{t}{\alpha} \left(E_{\alpha,1}(-z) + (\alpha - 1) E_{\alpha,2}(-z) \right). \quad (7)$$

The critical initial-displacement multiplier remains

$$n_{\text{crit}}(t, \xi) = E_{\alpha,1}(-z) + \frac{z}{\alpha} E_{\alpha,\alpha}(-z). \quad (8)$$

With (6)–(8), the Laplace-transformed Cauchy equations and all prescribed initial data are satisfied.

Proof. Equations (6) and (7) follow from (4)–(5) and Theorem 1, taking β and 2, respectively. For (8), decompose

$$\frac{s^{2\alpha-1} + 2as^{\alpha-1}}{(s^\alpha + a)^2} = \frac{s^{\alpha-1}}{s^\alpha + a} + \frac{as^{\alpha-1}}{(s^\alpha + a)^2}.$$

Its inverse is $E_{\alpha,1}(-z) + zE_{\alpha,\alpha+1}^2(-z)$, and (2) with $\beta = \alpha + 1$ gives $E_{\alpha,\alpha+1}^2 = E_{\alpha,\alpha}/\alpha$. The Laplace transforms are therefore exactly the rational multipliers derived from the Cauchy equations. $\square$

$$(6.1) \quad \lambda_{\pm} = \begin{cases} \frac{\mu}{2} \pm \sqrt{\frac{\mu^2}{4} - 1} & , \mu \geq 2 \end{cases}$$

as well as the kernel functions $M_{\sigma,n}(t, x|\alpha, \beta, \mu)$, $N_{\sigma,n}(t, x|\alpha, \mu)$ and $J_{\sigma,n}(t, x|\alpha, \mu)$, introduced as below, will be required on the proof of Theorem 6.3 Theorem 6.4 and elsewhere.

Notation 6.1 (Kernel functions underlying to Cauchy Problems 2.1 & 2.2).

$$\begin{aligned} M_{\sigma,n}(t, x|\alpha, \beta, \mu) &= \\ &= \begin{cases} \frac{t^{\beta-1}}{(2\pi)^n} \int_{\mathbb{R}^n} \frac{\lambda_+ E_{\alpha,\beta}(-\lambda_+|\xi|^{\sigma}t^\alpha) - \lambda_- E_{\alpha,\beta}(-\lambda_-|\xi|^{\sigma}t^\alpha)}{\lambda_+ - \lambda_-} e^{ix \cdot \xi} d\xi & , \mu \neq 2 \\ \frac{t^{\beta-1}}{\alpha(2\pi)^n} \int_{\mathbb{R}^n} (E_{\alpha,\beta-1}(-|\xi|^{\sigma}t^\alpha) - (1 + \alpha - \beta)E_{\alpha,\beta}(-|\xi|^{\sigma}t^\alpha)) e^{ix \cdot \xi} d\xi & , \mu = 2. \end{cases} \\ N_{\sigma,n}(t, x|\alpha, \mu) &= \\ &= \begin{cases} \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \left(\frac{E_{\alpha,1}(-\lambda_+|\xi|^{\sigma}t^\alpha)}{1 - (\lambda_+)^2} + \frac{E_{\alpha,1}(-\lambda_-|\xi|^{\sigma}t^\alpha)}{1 - (\lambda_-)^2} \right) e^{ix \cdot \xi} d\xi & , \mu \neq 2 \\ \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \left(E_{\alpha,1}(-|\xi|^{\sigma}t^\alpha) + \frac{|\xi|^{\sigma}t^\alpha}{\alpha} E_{\alpha,\alpha}(-|\xi|^{\sigma}t^\alpha) \right) e^{ix \cdot \xi} d\xi & , \mu = 2. \end{cases} \end{aligned}$$

Notation 6.2 (Kernel function underlying to Cauchy Problem 2.2).

$$\begin{aligned} J_{\sigma,n}(t, x|\alpha, \mu) &= \\ &= \begin{cases} \frac{t}{(2\pi)^n} \int_{\mathbb{R}^n} \left(\frac{\lambda_+ E_{\alpha,2}(-\lambda_+|\xi|^{\sigma}t^\alpha) - \lambda_- E_{\alpha,2}(-\lambda_-|\xi|^{\sigma}t^\alpha)}{\lambda_+ - \lambda_-} \right) e^{ix \cdot \xi} d\xi & , \mu \neq 2 \\ \frac{t}{\alpha(2\pi)^n} \int_{\mathbb{R}^n} (E_{\alpha,1}(-|\xi|^{\sigma}t^\alpha) - (\alpha - 1)E_{\alpha,2}(-|\xi|^{\sigma}t^\alpha)) e^{ix \cdot \xi} d\xi & , \mu = 2. \end{cases} \end{aligned}$$

6.2. Solution of the Cauchy Problem 2.1

Theorem 6.3. Let $0 < \alpha \leq \frac{1}{2}$, $2\alpha \leq \beta < 2\alpha + 1$, $\sigma > 0$ and $\mu > 0$ be given. If $u_0 \in \mathcal{S}(\mathbb{R}^n)$ and $f(t, \cdot) \in \mathcal{S}(\mathbb{R}^n)$ for every $t > 0$ then $u : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined

Figure 2: The critical kernels as printed on page 23. The M and J lines have the erroneous signs; the N line is correct.

4 Direct counterexample to Theorem 6.4

Theorem 2 (Failure of the printed initial condition). Fix $1/2 < \alpha < 1$, $\sigma > 0$, and $\mu = 2$. Let $u_0 = f = 0$ and choose nonzero $u_1 \in \mathcal{S}(\mathbb{R}^n)$ with $\hat{u}_1 \in C_c^\infty(\mathbb{R}^n)$. The function constructed from the printed kernel $J_{\sigma,n}$ in Theorem 6.4 satisfies

$$\partial_t u(0, \cdot) = \frac{2 - \alpha}{\alpha} u_1 \neq u_1.$$

It is therefore not a solution of Cauchy Problem 2.2.

Proof. The printed Fourier multiplier is

$$j_{\text{print}}(t, \xi) = \frac{t}{\alpha} (E_{\alpha,1}(-z) - (\alpha - 1)E_{\alpha,2}(-z)).$$

Since $E_{\alpha,1}(0) = E_{\alpha,2}(0) = 1$,

$$\frac{j_{\text{print}}(t, \xi)}{t} \longrightarrow \frac{1 - (\alpha - 1)}{\alpha} = \frac{2 - \alpha}{\alpha}.$$

The convergence is in C^∞ on the compact support of $\widehat{u}_1$, by the entire power series. Fourier inversion therefore gives convergence in $\mathcal{S}$ after multiplication by $\widehat{u}_1$. Since $u(0) = 0$, the asserted initial derivative follows. The multiplier (7) instead has limit $[1 + (\alpha - 1)]/\alpha = 1$ and restores the datum. $\square$

For the forcing kernel, the printed and correct formulas coincide only when $\beta = \alpha + 1$; equation (3) gives their exact difference otherwise. For the velocity kernel they coincide only when $\alpha = 1$. Thus the classical repeated-root case survives, while the genuinely fractional critical case does not.

5 Scope

This correction concerns the current arXiv v2. It refutes the critical $\mu = 2$ part of Theorem 6.4 as printed and the forced critical part of Theorems 6.3–6.4 outside the exceptional parameter above. It does not claim that the paper’s later dispersive or Strichartz inequalities fail: many of those arguments use absolute-value estimates and may be insensitive to the sign once the solution kernels are corrected. Those downstream claims require a separate audit.

A bounded full-source search and exact web searches found no erratum or later correction. Novelty is therefore provisional pending author or specialist review.

References

- [1] N. Faustino and J. Marques, *Structurally damped σ -evolution equations with power-law memory*, arXiv:2212.10463v2 (2023).